

Problem A. Concentration inequality

In this problem, we will derive a concentration inequality that is finer than Hoeffding's inequality.

Let $X_1, \dots, X_m$ be independent random variables drawn according to some distribution $\mathcal{D}$ with mean p and support included in $[0, 1]$. Let $q \in [0, 1]$ and define $\hat{p}$ by $\hat{p} = \frac{1}{m} \sum_{i=1}^m X_i$.

1. Use Markov's inequality to prove that for any $t > 0$,

$$\mathbb{P}[\hat{p} \geq q] \leq e^{-tmq} \prod_{i=1}^m \mathbb{E}[e^{tX_i}].$$

2. Show that

$$\mathbb{P}[\hat{p} \geq q] \leq [e^{-tq} (1 - p + e^tp)]^m$$

(Hint: you can show that $\forall x \in [0, 1], e^{tx} \leq 1 - x + e^tx$).

3. Choose $t > 0$ to minimize the right-hand side and prove the following inequality:

$$\mathbb{P}[\hat{p} \geq q] \leq e^{-mD(q||p)},$$

where $D(q||p) = q \log \frac{q}{p} + (1 - q) \log \frac{1-q}{1-p}$ is the binary relative entropy of p and q .

4. Show that this can be used to derive a bound that is finer than Hoeffding's inequality by choosing $q = p + \varepsilon$, for any $\varepsilon > 0$ (Hint: you can use Pinsker's inequality, which in this context gives $D(q||p) \geq 2|p - q|^2$).

(1) Recall that if Y is a nonnegative random variable and $a > 0$, then Markov's inequality states:

$$\mathbb{P}[Y \geq a] \leq \frac{\mathbb{E}[Y]}{a}.$$

Here, let

$$Y = e^{tm\hat{p}} \quad \text{and} \quad a = e^{tmq}.$$

Then,

$$\mathbb{P}(e^{tm\hat{p}} \geq e^{tmq}) \leq \frac{\mathbb{E}[e^{tm\hat{p}}]}{e^{tmq}}.$$

Since the event $\{e^{tm\hat{p}} \geq e^{tmq}\}$ is equivalent to $\{\hat{p} \geq q\}$, we have:

$$\mathbb{P}(\hat{p} \geq q) \leq e^{-tmq} \mathbb{E}[e^{tm\hat{p}}].$$

Notice that

$$\hat{p} = \frac{1}{m} \sum_{i=1}^m X_i \implies e^{tm\hat{p}} = e^{t \sum_{i=1}^m X_i}.$$

Since the X_i are independent, the expectation of the exponential of the sum factors as a product:

$$\mathbb{E} \left[e^{t \sum_{i=1}^m X_i} \right] = \prod_{i=1}^m \mathbb{E} \left[e^{t X_i} \right].$$

Thus, we obtain:

$$\mathbb{P}(\hat{p} \geq q) \leq e^{-tmq} \prod_{i=1}^m \mathbb{E} \left[e^{t X_i} \right].$$

(2) Define the function

$$f(x) = e^{tx}.$$

We compute its second derivative:

$$f''(x) = t^2 e^{tx}.$$

Since $t > 0$ and $e^{tx} > 0$ for all x , we have $f''(x) > 0$ for all x . This shows that $f(x)$ is a *convex* function.

For any $x \in [0, 1]$, we can express x as a convex combination of 0 and 1:

$$x = (1 - x) \cdot 0 + x \cdot 1.$$

Using the property of convex functions, we have:

$$f((1 - x) \cdot 0 + x \cdot 1) \leq (1 - x)f(0) + xf(1).$$

the inequality becomes

$$e^{tx} \leq (1 - x) \cdot 1 + x \cdot e^t = 1 - x + e^t x.$$

This completes the proof of hint.

Taking expectation with respect to X_i yields

$$\mathbb{E} \left[e^{t X_i} \right] \leq \mathbb{E} \left[1 - X_i + e^t X_i \right].$$

Because expectation is linear, we have

$$\mathbb{E} \left[1 - X_i + e^t X_i \right] = 1 - \mathbb{E}[X_i] + e^t \mathbb{E}[X_i] = 1 - p + e^t p.$$

Since the same bound holds for each X_i , we get

$$\prod_{i=1}^m \mathbb{E} \left[e^{t X_i} \right] \leq (1 - p + e^t p)^m.$$

Substituting this into our earlier bound gives

$$\mathbb{P}(\hat{p} \geq q) \leq e^{-tmq} (1 - p + e^t p)^m.$$

Since $e^{-tmq} = \left(e^{-tq}\right)^m$, we can write:

$$\mathbb{P}[\hat{p} \geq q] \leq \left[e^{-tq}(1 - p + e^t p)\right]^m.$$

This completes the proof.

(3)

Let

$$f(t) = e^{-tq}(1 - p + p e^t).$$

We wish to choose $t > 0$ so that $f(t)$ is minimized.

Define

$$\phi(t) = \ln f(t) = -tq + \ln(1 - p + p e^t).$$

Differentiating with respect to t gives

$$\phi'(t) = -q + \frac{p e^t}{1 - p + p e^t}.$$

Setting $\phi'(t) = 0$ yields

$$\frac{p e^t}{1 - p + p e^t} = q.$$

Solving for e^t :

$$p e^t = q(1 - p + p e^t) \implies p e^t = q(1 - p) + q p e^t.$$

Rearrange the terms to isolate e^t :

$$p e^t(1 - q) = q(1 - p) \implies e^t = \frac{q(1 - p)}{p(1 - q)}.$$

Thus, the minimizing value is

$$t^* = \ln\left(\frac{q(1 - p)}{p(1 - q)}\right).$$

Substitute $t = t^*$ into the bound:

$$f(t^*) = e^{-q t^*}(1 - p + p e^{t^*}).$$

Since

$$e^{t^*} = \frac{q(1 - p)}{p(1 - q)},$$

we have:

$$e^{-q t^*} = \left(\frac{q(1 - p)}{p(1 - q)}\right)^{-q},$$

and

$$1 - p + p e^{t^*} = 1 - p + p \frac{q(1 - p)}{p(1 - q)} = 1 - p + \frac{q(1 - p)}{1 - q} = (1 - p) \left(1 + \frac{q}{1 - q}\right) = \frac{1 - p}{1 - q}.$$

Thus,

$$f(t^*) = \left(\frac{q(1-p)}{p(1-q)} \right)^{-q} \frac{1-p}{1-q}.$$

Rewrite the expression and we obtain

$$f(t^*) = \left(\frac{p}{q} \right)^q (1-p)^{1-q} (1-q)^{-(1-q)} = \left(\frac{p}{q} \right)^q \left(\frac{1-p}{1-q} \right)^{1-q}.$$

Observing that

$$e^{-D(q||p)} = \exp \left[-q \log \frac{q}{p} - (1-q) \log \frac{1-q}{1-p} \right] = \left(\frac{p}{q} \right)^q \left(\frac{1-p}{1-q} \right)^{1-q},$$

we conclude that

$$f(t^*) = e^{-D(q||p)}.$$

Thus, the optimal bound becomes

$$\mathbb{P}[\hat{p} \geq q] \leq [f(t^*)]^m = e^{-m D(q||p)}.$$

(4) choose

$$q = p + \varepsilon \quad \text{for any } \varepsilon > 0.$$

Then the bound becomes

$$\mathbb{P}[\hat{p} \geq p + \varepsilon] \leq \exp \left(-m D(p + \varepsilon || p) \right).$$

Pinsker's inequality, tells us that in this context

$$D(p + \varepsilon || p) \geq 2\varepsilon^2.$$

Substituting this into our bound, we obtain

$$\mathbb{P}[\hat{p} \geq p + \varepsilon] \leq e^{-2m\varepsilon^2}.$$

This bound is at least as strong (and under many circumstances finer) than Hoeffding's inequality. because:

- The binary relative entropy $D(p + \varepsilon || p)$ depends on the specific values of p and $p + \varepsilon$, thereby capturing the true large-deviation rate. In contrast, Hoeffding's inequality provides a uniform *worst-case* bound over all distributions with support in $[0, 1]$.
- Pinsker's inequality gives the lower bound $D(p + \varepsilon || p) \geq 2\varepsilon^2$, but $D(p + \varepsilon || p)$ is typically *greater* than $2\varepsilon^2$ unless $p = 1/2$ (or in the limit as $\varepsilon \rightarrow 0$). For example, when p is near 0 or 1, the divergence $D(p + \varepsilon || p)$ can be significantly larger than $2\varepsilon^2$, which implies a much faster exponential decay for the tail probability.

Problem B. PAC-Learning

1. **Concentric spheres.** Let $\mathcal{X} = \mathbb{R}^3$ and consider the set of concepts of the form

$$c = \{(x, y, z) : x^2 + y^2 + z^2 \leq r^2\} \quad \text{for some } r > 0.$$

Show that this class can be (ε, δ) -PAC-learned from training data of size

$$m \geq \frac{1}{\varepsilon} \log\left(\frac{1}{\delta}\right).$$

You should carefully justify all steps of your proof. (For this problem, you should seek a proof similar to one presented in class for rectangles in the plane and not leverage a more powerful or more general theorem.)

We consider the concept class

$$\mathcal{C} = \{c_r : c_r = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 \leq r^2\}, r > 0\},$$

and assume the target concept is

$$c = c_{r_0} = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 \leq r_0^2\},$$

for some unknown $r_0 > 0$. The distribution D over $\mathbb{R}^3$ is arbitrary.

Then, we can use this Algorithm: Given m i.i.d. training examples

$$S = \{(x_1, y_1), (x_2, y_2), \dots, (x_m, y_m)\},$$

where $x_i \sim D$ and

$$y_i = \mathbf{1}_{x_i \in c},$$

the algorithm outputs the hypothesis

$$h = \{x \in \mathbb{R}^3 : \|x\| \leq r_+\},$$

where

$$r_+ = \max\{\|x_i\| : y_i = 1\}$$

is the maximum distance from the origin among all positive examples.

The correctness is that since for any positive example we have $\|x_i\| \leq r_0$, it follows that

$$r_+ \leq r_0,$$

so that

$$h \subseteq c.$$

Thus, the hypothesis h never labels a negative example as positive and the only possible error is failing to cover some positive examples. In particular, the error is

$$\text{error}_D(h) = \Pr_{x \sim D} [x \in c \setminus h].$$

PAC Guarantee:

Fix $\varepsilon, \delta \in (0, 1)$. We wish to show that if

$$m \geq \frac{1}{\varepsilon} \ln \left(\frac{1}{\delta} \right),$$

then

$$\Pr [\text{error}_D(h) > \varepsilon] \leq \delta.$$

The proof proceeds similarly to the one for axis-aligned rectangles in the plane:

Since $h \subseteq c$, the error comes solely from points in the annulus (the outer shell) of c . Choose a radius r' with $0 \leq r' < r_0$ so that the outer region

$$A = \{x \in c : r' < \|x\| \leq r_0\}$$

has probability

$$\Pr_D[A] \geq \varepsilon.$$

(If such an A does not exist, namely, if $\Pr_D[c] \leq \varepsilon$ then for any hypothesis $h \subseteq c$ the error satisfies $\text{error}_D(h) \leq \Pr_D[c] \leq \varepsilon$, and the claim holds trivially.)

Notice that if at least one positive training example lies in A , then the maximum radius r_+ will be at least r' , and therefore

$$h = \{x : \|x\| \leq r_+\} \supseteq \{x : \|x\| \leq r'\}.$$

Thus,

$$c \setminus h \subseteq c \setminus \{x : \|x\| \leq r'\} = A,$$

and consequently,

$$\text{error}_D(h) \leq \Pr_D[A] \leq \varepsilon.$$

In other words, if a positive training example falls in A , then the error is at most ε .

If $\text{error}_D(h) > \varepsilon$, then no positive training example in S fell in A . For each training example, the probability (unconditionally) that it lands in A is at least ε . Therefore, the probability that a single training example does *not* fall in A is at most $1 - \varepsilon$. Since the examples are independent, the probability that none of the m examples falls in A is bounded by

$$(1 - \varepsilon)^m \leq e^{-m\varepsilon}.$$

To guarantee that

$$\Pr [\text{error}_D(h) > \varepsilon] \leq e^{-m\varepsilon} \leq \delta,$$

it suffices to choose m so that

$$e^{-m\varepsilon} \leq \delta.$$

Taking logarithms we have

$$-m\varepsilon \leq \ln \delta,$$

or equivalently,

$$m \geq \frac{1}{\varepsilon} \ln \left(\frac{1}{\delta} \right).$$

This completes the proof □

Problem C. Rademacher complexity

1. Let $\mathcal{H}_1$ and $\mathcal{H}_2$ be two families of functions mapping $\mathcal{X}$ to $[0, 1]$ and let S be a sample of size m .

- (a) Show that for any sample S ,

$$\widehat{\mathfrak{R}}_S(\mathcal{H}_1 + \mathcal{H}_2) = \widehat{\mathfrak{R}}_S(\mathcal{H}_1) + \widehat{\mathfrak{R}}_S(\mathcal{H}_2).$$

- (b) Show that the empirical Rademacher complexity of $\mathcal{H}$ for any sample S of size m can be bounded as follows:

$$\widehat{\mathfrak{R}}_S(\mathcal{H}_1 \mathcal{H}_2) \leq \frac{3}{2} \left[\widehat{\mathfrak{R}}_S(\mathcal{H}_1) + \widehat{\mathfrak{R}}_S(\mathcal{H}_2) \right].$$

Hint: use the identity $x_1 x_2 = \frac{1}{4} [(x_1 + x_2)^2 - (x_1 - x_2)^2]$ valid for all $x_1, x_2 \in \mathbb{R}$ and Talagrand's contraction lemma.

2. Let $\mathcal{E}$ be the family of functions defined over $\mathcal{X}$ by $\mathcal{E} = \{x \mapsto \mathbb{I}_{x=z} : z \in \mathcal{X}\}$.

- (a) Let S be a sample of size m with distinct points. Prove the inequality $\widehat{\mathfrak{R}}_S(\mathcal{E}) \leq \frac{1}{m}$.
- (b) Let S be a sample of size m with a single point repeated m times. Prove the equality

$$\widehat{\mathfrak{R}}_S(\mathcal{E}) = \frac{1}{2m} \mathbb{E}_{\sigma} \left[\left| \sum_{i=1}^m \sigma_i \right| \right].$$

Show that $\widehat{\mathfrak{R}}_S(\mathcal{E}) \leq \frac{1}{2\sqrt{m}}$.

- (c) 1[bonus question] Let n^* denote the largest number of repetitions of any point in the sample. Give an upper bound on $\widehat{\mathfrak{R}}_S(\mathcal{E})$ in terms of n^* .

(1) For the class $\mathcal{H}_1 + \mathcal{H}_2$, we have

$$\begin{aligned}\widehat{\mathfrak{R}}_S(\mathcal{H}_1 + \mathcal{H}_2) &= \frac{1}{m} \mathbb{E}_\sigma \left[\sup_{h_1 \in \mathcal{H}_1, h_2 \in \mathcal{H}_2} \sum_{i=1}^m \sigma_i (h_1(x_i) + h_2(x_i)) \right] \\ &= \frac{1}{m} \mathbb{E}_\sigma \left[\sup_{h_1 \in \mathcal{H}_1} \sum_{i=1}^m \sigma_i h_1(x_i) + \sup_{h_2 \in \mathcal{H}_2} \sum_{i=1}^m \sigma_i h_2(x_i) \right] \\ &= \widehat{\mathfrak{R}}_S(\mathcal{H}_1) + \widehat{\mathfrak{R}}_S(\mathcal{H}_2).\end{aligned}$$

This completes the proof for part (a).

Let us now consider the class $\mathcal{H}_1 \mathcal{H}_2 = \{h_1 \cdot h_2 : h_1 \in \mathcal{H}_1, h_2 \in \mathcal{H}_2\}$. By definition,

$$\widehat{\mathfrak{R}}_S(\mathcal{H}_1 \mathcal{H}_2) = \frac{1}{m} \mathbb{E}_\sigma \left[\sup_{h_1 \in \mathcal{H}_1, h_2 \in \mathcal{H}_2} \sum_{i=1}^m \sigma_i h_1(x_i) h_2(x_i) \right].$$

For any real numbers x_1, x_2 the following identity holds:

$$x_1 x_2 = \frac{1}{4} \left[(x_1 + x_2)^2 - (x_1 - x_2)^2 \right].$$

Hence, replacing $x_1 = r_i := h_1(x_i)$ and $x_2 = s_i := h_2(x_i)$ (noting that since $h_1(x_i), h_2(x_i) \in [0, 1]$ the sums $h_1(x_i) + h_2(x_i) \in [0, 2]$ and differences $h_1(x_i) - h_2(x_i) \in [-1, 1]$) we obtain

$$h_1(x_i) h_2(x_i) = \frac{1}{4} \left[(h_1(x_i) + h_2(x_i))^2 - (h_1(x_i) - h_2(x_i))^2 \right].$$

Thus, for any fixed choice of Rademacher signs,

$$\sum_{i=1}^m \sigma_i h_1(x_i) h_2(x_i) = \frac{1}{4} \sum_{i=1}^m \sigma_i \left[(h_1(x_i) + h_2(x_i))^2 - (h_1(x_i) - h_2(x_i))^2 \right].$$

Taking the supremum over h_1 and h_2 and then the expectation yields

$$\widehat{\mathfrak{R}}_S(\mathcal{H}_1 \mathcal{H}_2) \leq \frac{1}{4} \left[\widehat{\mathfrak{R}}_S(\{(h_1 + h_2)^2\}) + \widehat{\mathfrak{R}}_S(\{(h_1 - h_2)^2\}) \right].$$

We now use Talagrand's contraction lemma. Recall that if $\phi: \mathbb{R} \rightarrow \mathbb{R}$ is L -Lipschitz with $\phi(0) = 0$, then

$$\widehat{\mathfrak{R}}_S(\phi \circ \mathcal{F}) \leq L \widehat{\mathfrak{R}}_S(\mathcal{F}).$$

For the Class $\{(h_1 + h_2)^2\}$: Since $h_1(x) + h_2(x) \in [0, 2]$, the function $\phi(x) = x^2$ is 4-Lipschitz on $[0, 2]$ (because for $x, y \in [0, 2]$,

$$|x^2 - y^2| = |x - y||x + y| \leq 4|x - y|).$$

Moreover, $\phi(0) = 0$. Thus, by the contraction lemma,

$$\widehat{\mathfrak{R}}_S(\{(h_1 + h_2)^2\}) \leq 4 \widehat{\mathfrak{R}}_S(\{h_1 + h_2\}).$$

Using part (a), we have

$$\widehat{\mathfrak{R}}_S(\{h_1 + h_2\}) = \widehat{\mathfrak{R}}_S(\mathcal{H}_1) + \widehat{\mathfrak{R}}_S(\mathcal{H}_2).$$

Thus,

$$\widehat{\mathfrak{R}}_S(\{(h_1 + h_2)^2\}) \leq 4 \left[\widehat{\mathfrak{R}}_S(\mathcal{H}_1) + \widehat{\mathfrak{R}}_S(\mathcal{H}_2) \right].$$

For the Class $\{(h_1 - h_2)^2\}$: Similarly, $h_1(x) - h_2(x) \in [-1, 1]$ and on $[-1, 1]$ the function $\psi(x) = x^2$ is 2-Lipschitz (since $|x^2 - y^2| \leq 2|x - y|$ for all $x, y \in [-1, 1]$ and $\psi(0) = 0$). Hence,

$$\widehat{\mathfrak{R}}_S(\{(h_1 - h_2)^2\}) \leq 2 \widehat{\mathfrak{R}}_S(\{h_1 - h_2\}) = 2 \left[\widehat{\mathfrak{R}}_S(\mathcal{H}_1) + \widehat{\mathfrak{R}}_S(\mathcal{H}_2) \right],$$

where in the last step we again used part (a) (noting that for any pair, $(h_1 - h_2)$ belongs to the sum of $\mathcal{H}_1$ and $-\mathcal{H}_2$; since the Rademacher complexity is invariant under sign changes, the same bound holds).

Plugging these estimates back into our bound for $\widehat{\mathfrak{R}}_S(\mathcal{H}_1 \mathcal{H}_2)$ we obtain

$$\begin{aligned} \widehat{\mathfrak{R}}_S(\mathcal{H}_1 \mathcal{H}_2) &\leq \frac{1}{4} \left[4 \left(\widehat{\mathfrak{R}}_S(\mathcal{H}_1) + \widehat{\mathfrak{R}}_S(\mathcal{H}_2) \right) + 2 \left(\widehat{\mathfrak{R}}_S(\mathcal{H}_1) + \widehat{\mathfrak{R}}_S(\mathcal{H}_2) \right) \right] \\ &= \frac{1}{4} \cdot 6 \left(\widehat{\mathfrak{R}}_S(\mathcal{H}_1) + \widehat{\mathfrak{R}}_S(\mathcal{H}_2) \right) = \frac{3}{2} \left(\widehat{\mathfrak{R}}_S(\mathcal{H}_1) + \widehat{\mathfrak{R}}_S(\mathcal{H}_2) \right). \end{aligned}$$

This completes the proof for part (b).

(2)

Proof:

For the class $\mathcal{E}$, each function is of the form

$$f_z(x) = \mathbf{1}_{\{x=z\}}, \quad \text{for some } z \in \mathcal{X}.$$

Consider a fixed sample $S = \{x_1, \dots, x_m\}$ where all x_i are distinct. There are two types of functions in $\mathcal{E}$:

1. If $z \notin S$, then for all i we have $f_z(x_i) = 0$ and thus

$$\sum_{i=1}^m \sigma_i f_z(x_i) = 0.$$

2. If $z = x_j$ for some $j \in \{1, \dots, m\}$, then

$$f_z(x_i) = \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{otherwise.} \end{cases}$$

Hence,

$$\sum_{i=1}^m \sigma_i f_z(x_i) = \sigma_j.$$

Thus, the supremum

$$\sup_{f \in \mathcal{E}} \sum_{i=1}^m \sigma_i f(x_i)$$

is achieved by choosing an index j for which $\sigma_j = +1$ if at least one exists; otherwise, if all $\sigma_i = -1$, one may choose a function corresponding to some $z \notin S$ (which gives sum 0) to obtain a larger value. In summary, we have

$$\sup_{f \in \mathcal{E}} \sum_{i=1}^m \sigma_i f(x_i) = \max \left\{ 0, \max_{1 \leq j \leq m} \sigma_j \right\}.$$

Since each σ_j takes values in $\{-1, +1\}$, the above supremum equals:

$$\sup_{f \in \mathcal{E}} \sum_{i=1}^m \sigma_i f(x_i) = \begin{cases} 1, & \text{if there exists some } j \text{ with } \sigma_j = +1; \\ 0, & \text{if } \sigma_i = -1 \text{ for all } i. \end{cases}$$

Therefore, by taking expectation over the random signs, we observe that

$$\mathbb{E}_\sigma \left[\sup_{f \in \mathcal{E}} \sum_{i=1}^m \sigma_i f(x_i) \right] = (1 - 2^{-m}) \cdot 1 + 2^{-m} \cdot 0 = 1 - 2^{-m}.$$

Finally, by the definition of empirical Rademacher complexity we have

$$\widehat{\mathfrak{R}}_S(\mathcal{E}) = \frac{1}{m} (1 - 2^{-m}) \leq \frac{1}{m},$$

since $1 - 2^{-m} \leq 1$ for all $m \geq 1$.

This completes the proof.

(b) Since every function in $\mathcal{E}$ is of the form

$$f_z(x) = \mathbf{1}_{\{x=z\}} \quad \text{for some } z \in \mathcal{X},$$

we examine its action on the sample S .

Because S contains only the single point x , we have: - If $z = x$: then $f_x(x) = 1$ so that for every i ,

$$f_x(x_i) = 1,$$

and therefore

$$\sum_{i=1}^m \sigma_i f_x(x_i) = \sum_{i=1}^m \sigma_i.$$

- For any $z \neq x$: then $f_z(x) = 0$ so that

$$f_z(x_i) = 0 \quad \text{for all } i,$$

leading to

$$\sum_{i=1}^m \sigma_i f_z(x_i) = 0.$$

Thus, the supremum over $\mathcal{E}$ is

$$\sup_{f \in \mathcal{E}} \sum_{i=1}^m \sigma_i f(x_i) = \max \left\{ \sum_{i=1}^m \sigma_i, 0 \right\},$$

which is the positive part of the sum. We denote it by

$$\left(\sum_{i=1}^m \sigma_i \right)_+.$$

Thus, the empirical Rademacher complexity becomes

$$\hat{\mathfrak{R}}_S(\mathcal{E}) = \frac{1}{m} \mathbb{E}_{\boldsymbol{\sigma}} \left[\left(\sum_{i=1}^m \sigma_i \right)_+ \right].$$

Because the Rademacher distribution is symmetric, the positive part and the negative part are balanced. In fact, note that

$$\left| \sum_{i=1}^m \sigma_i \right| = \left(\sum_{i=1}^m \sigma_i \right)_+ + \left(- \sum_{i=1}^m \sigma_i \right)_+ = 2 \left(\sum_{i=1}^m \sigma_i \right)_+,$$

where the last equality holds by symmetry (i.e. $\mathbb{E}[(\sum \sigma_i)_+] = \mathbb{E}[(-\sum \sigma_i)_+]$). Taking expectation, we obtain

$$\mathbb{E}_{\boldsymbol{\sigma}} \left[\left(\sum_{i=1}^m \sigma_i \right)_+ \right] = \frac{1}{2} \mathbb{E}_{\boldsymbol{\sigma}} \left[\left| \sum_{i=1}^m \sigma_i \right| \right].$$

Substituting back, we have

$$\hat{\mathfrak{R}}_S(\mathcal{E}) = \frac{1}{2m} \mathbb{E}_{\boldsymbol{\sigma}} \left[\left| \sum_{i=1}^m \sigma_i \right| \right].$$

To bound $\mathbb{E} \left[\left| \sum_{i=1}^m \sigma_i \right| \right]$, we can use the following standard calculation. First, notice that

$$\mathbb{E} \left[\left(\sum_{i=1}^m \sigma_i \right)^2 \right] = \sum_{i=1}^m \mathbb{E}[\sigma_i^2] = m,$$

since the σ_i are independent with $\sigma_i^2 = 1$ almost surely. Now, by Jensen's inequality, we have

$$\mathbb{E} \left[\left| \sum_{i=1}^m \sigma_i \right| \right] \leq \sqrt{\mathbb{E} \left[\left(\sum_{i=1}^m \sigma_i \right)^2 \right]} = \sqrt{m}.$$

Thus,

$$\hat{\mathfrak{R}}_S(\mathcal{E}) \leq \frac{1}{2m} \sqrt{m} = \frac{1}{2\sqrt{m}}.$$

(c):

Let $S = \{x_1, \dots, x_m\}$ be a sample containing k distinct points $z_1, \dots, z_k$, where point z_j appears n_j times. We have $n^* = \max_{j=1, \dots, k} n_j$ and $\sum_{j=1}^k n_j = m$.

For any function $e_z \in \mathcal{E}$ where $e_z(x) = \mathbf{1}_{\{x=z\}}$, we have:

$$\sum_{i=1}^m \sigma_i e_z(x_i) = \begin{cases} \sum_{i: x_i=z_j} \sigma_i & \text{if } z = z_j \text{ for some } j \in \{1, \dots, k\} \\ 0 & \text{if } z \notin \{z_1, \dots, z_k\} \end{cases}$$

Therefore:

$$\sup_{e \in \mathcal{E}} \sum_{i=1}^m \sigma_i e(x_i) = \max \left\{ 0, \max_{j=1, \dots, k} \sum_{i: x_i=z_j} \sigma_i \right\}$$

Constructing the finite set, for each $j \in \{1, \dots, k\}$, define the vector $x^{(j)} \in \mathbb{R}^m$ by:

$$x_i^{(j)} = \begin{cases} 1 & \text{if the } i\text{-th sample point equals } z_j \\ 0 & \text{otherwise} \end{cases}$$

Also define $x^{(0)} = 0 \in \mathbb{R}^m$ (the zero vector).

Let $A = \{x^{(0)}, x^{(1)}, \dots, x^{(k)}\} \subset \mathbb{R}^m$. Then:

$$\sup_{e \in \mathcal{E}} \sum_{i=1}^m \sigma_i e(x_i) = \sup_{x \in A} \sum_{i=1}^m \sigma_i x_i$$

Computing the parameters for Massart's Lemma:

- The set A is finite with $|A| = k + 1 \leq m + 1$
- For $j \in \{1, \dots, k\}$: $\|x^{(j)}\|_2 = \sqrt{n_j} \leq \sqrt{n^*}$
- For $j = 0$: $\|x^{(0)}\|_2 = 0$
- Therefore: $R = \max_{x \in A} \|x\|_2 = \max_{j=1, \dots, k} \sqrt{n_j} = \sqrt{n^*}$

By Massart's Lemma, for the finite set $A \subseteq \mathbb{R}^m$ with $R = \max_{x \in A} \|x\|_2 = \sqrt{n^*}$:

$$\mathbb{E}_\sigma \left[\frac{1}{m} \sup_{x \in A} \sum_{i=1}^m \sigma_i x_i \right] \leq \frac{R \sqrt{2 \log |A|}}{m} = \frac{\sqrt{n^*} \cdot \sqrt{2 \log(k+1)}}{m}$$

Since $k \leq m$, we have $\log(k+1) \leq \log(m+1)$. Therefore:

$$\widehat{\mathfrak{R}}_S(\mathcal{E}) = \mathbb{E}_\sigma \left[\frac{1}{m} \sup_{e \in \mathcal{E}} \sum_{i=1}^m \sigma_i e(x_i) \right] \leq \frac{\sqrt{2n^* \log(m+1)}}{m}$$